PIC12/14/16 Integrated Development System

Integrated Development Environment

MR.PIC-IDE™

USER'S MANUAL

COMFILE TECHNOLOGY

www.comfile.co.kr

WARNING...

Follow the precautions listed below to avoid permanent damage to the MR.PIC.
If the target board has its own power supply, follow the power-up and power-down procedures as shown below;

1. Always use a grounding strap to prevent damage resulting from electrostatic discharge(ESD).

2. Power-Up Procedure.
    1) Connect the probe to the target board
    2) Turn on MR.PIC
       (if MR.PIC has the power supply)
       Turn on target board
       (if target board has the power supply)

3. Power-Down Procedure
    1) Turn-off the target board
       (if target board has the power supply)
    2) Turn-off MR.PIC
       (if MR.PIC has the power supply)
    3) Remove the probe and the cable

About MR.PIC

MR.PIC is a real-time in-circuit emulator for MICROCHIP PIC12/14/16 microcontroller family. MR.PIC has two functoin :

- Real time In-circuit-emulator
- Universal programmer

MR.PIC supports all PIC16C5X device ; these include the PIC16C52, 54, 55, 56, 57, 58. (It allows you to run PIC16C5X programs in-circuit at the maximum speed of 20MHZ.)

Supports PIC16CXX device ; these include the PIC16C62, 63, 64, 65, PIC16C554, 556, 558, PIC16C72, 73, 74 (It allows you to run PIC16CXX programs in-circuit at the maximum speed of 10MHZ.)

Supports PIC12CXXX device ; these include the PIC12C508, 509, 671, 672 (It allows you to run PIC16CXX programs in-circuit at the maximum speed of 20MHZ.)

Another devices can operate with related "ME chip". The followings are examples of probes which possibly can use :

PIC16C711-ME ; support PIC16C71, 710, 711
PIC16CR84-ME ; support PIC16F83, F84, C84
PIC16C622-ME ; support PIC16C620, 621, 622

MICROCHIP continuously support ME chip for supporting PIC16CXX family which will be announce later.

About MR.PIC-IDE

MR.PIC-IDE is Integrated Development Environment software for MR.PIC.

MR.PIC-IDE is an easy-to-use GUI (graphic user interface) under MS WINDOWS (WINDOWS 3.1 or WINDOWS 95) and Integrated Development Environment (include EDITOR, DEBUGER, ASSEMBLER, PROGRAMMER S/W). You can source program editing, assemble (or compile), download, step through code, run, halt, set breakpoints (unlimited numbers), and modify registers.

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CHAPTER 1

INTRODUCTION

OVERVIEW

Congratulations ! You have selected the powerful development tool, the MR.PIC. The MR.PIC is carefully engineered to provide powerful features and low-cost to shorten your development time for PIC12CXXX, PIC16C5X, PIC16CXX based products.

KEY FEATURES

1. Integrated Development Environment (Include EDITOR, Debug ging feature)

1. Multiple windows can be viewed and the contents of Source Window (Editable), Special Register, Variable Register, Watch windows, Data memory (Register File), Program Memory Map and Error Windows.

1. Program Run, Single-Step Run, Break, and Edit the contents of Data Memory (Special Register, Variable Register)

1. Unlimited set break points can be set in Source list window.

1. Host interface is parallel port (Printer port)

1. Source level debugging.

1. Support MPLAB-C (MICROCHIP) and MP-C (BYTECRAFT)

1. And more ; Watch windows, Animate (Slow Run), Bit helps...

DEFAULT SUPPORTED DEVICES

PIC12CXXX FAMILY
PIC12C508, PIC12C509, PIC12C671, PIC12C672

PIC16C5X FAMILY
PIC16C52,PIC16C54,PIC16C55,PIC16C56,PIC16C57,PIC16C58

PIC16C6X FAMILY
PIC16C62, PIC16C63, PIC16C64, PIC16C65, PIC16C66, PIC16C67

PIC16C7X FAMILY
PIC16C72, PIC16C73, PIC16C74, PIC16C76, PIC16C77

PIC16C55X FAMILY
PIC16C554, PIC16C556, PIC16C558

PIC14C000 FAMILY
PIC14C000

OBJECT FORMATS

The MR.PIC supports the object (".COD" format) produced by the MICROCHIP MPASM assembler or MPLAB-C complier.

HARDWARE SPECIFICATIONS

MR.PIC

Dimensions	22.7cm x 18.1cm x 5.6cm
Target Oscillator	DC ~ 20MHz (selectable dip s/w 32KHz, 1MHz, 4MHz, 8MHz, 10MHz, 16MHz, 20MHz)
Host Interface	Printer port Parallel Interface
Power Supply Voltage	±±15VDC or ±±12VAC
Power Supply Current	500mA typical (at ±±9VDC)
Operating Temperature	20 degrees, ±± 10 degrees C
Operating Humidity	10~90% RH(non-condensing)

REQUIREMENTS AND RESTRICTIONS

Minimum Hardware requirement is IBM PC (or compatible) 486-based machine at 66MHz with 8M RAM, hard disk drive (with at least 1Mbytes available), 3.5 floppy disk drive, VGA video display, and a mouse, and keyboard.

We recommend a PENTIUM machine (66MHz) and 32M byte of RAM, which will provide enhanced overall performance.

Minimum Software Operating Systems

MS-DOS V5.0 with MS-WINDOWS 3.1 or MS-WINDOWS 95

Additional Required Items Not Supplied with the Support Package

A power source (±18VDC) for MR.PIC. This can be a laboratory power supply with supply current of 500mA. A wall-transfer-type power supply, can also be used ; however, a laboratory-type power supply is recommended, especially when using circuits sensitive to noise, such as Analog-to-Digital converters.

Restrictions

This emulator is designed for emulation purposes only and cannot be used in a stand alone mode.

The RC oscillator emulation and the external CRYSTAL oscillator emulation is not supported.

CHAPTER 2

SETUP & INSTALLATION

Installing The MR.PIC-IDE (Software)

MR.PIC contains a MR.PIC-IDE diskette. First, insert this diskette into the drive, then do following steps.

1. Start Windows 3.1 or Window95
2. Select the "Run" command from the "File" menu under Microsoft Windows 3.1 "Program Manager". (If Windows 95, select the "Run" command from the "Start" Menu at the bottom of the screen.)
3. Type "a:\setup" and press ENTER. (Type "b:setup" if drive B is used.) The setup program will copy the files into the hard disk drive and, creating a MR.PIC-IDE icon in the windows environment. After the installation is completed, you can move the MR.PIC-IDE icon into any program group of your choice.
4. Remove diskette and store in a safe place when you are done.



Figure 2-1. Setup MR.PIC-LAB setup

Note:
Modification of the "win.ini", "autoexec.bat", or "config.sys" files is not required.
In the installation procedure, the following files are copied to "windows directory" or "windows\system" directory.

MRPIC directory :	windows\system directory
MRPICIDE.EXE	LABRPV.DLL
MRPICPGM.EXE	LABDLG.VBX
MPASM.EXE	PROLAB.DLL
MPASMWIN.EXE	LABGAUGE.VBX
SAMPLE55.ASM	LABTHREE.VBX
SAMPLE64.ASM	COMMDLG.DLL
SAMPLE74.ASM	LABCOMM.VBX
	VBRUN300.DLL

We recommend that windows screen is setup in the Super VGA mode (800*600, 16 color). If MR.PIC-IDE is too slow, setup the screen to the Super VGA mode.
You should close other windows the other applications before the execution of MR.PIC-IDE.

Installing The MR.PIC (Hardware)

MR.PIC hardware install

1. You connect the cable to the PC printer port, and the MR.PIC.
2. You should connecting to a power supply.

If your power supply allows voltage adjustment, do the following:

- Turn the power supply on and adjust it to +18 volts.
- Set it for at least 500mA, if there's a current-limiting adjustment.

3. You put MR.PIC to a target board.

The Figure below shows MR.PIC after installation.

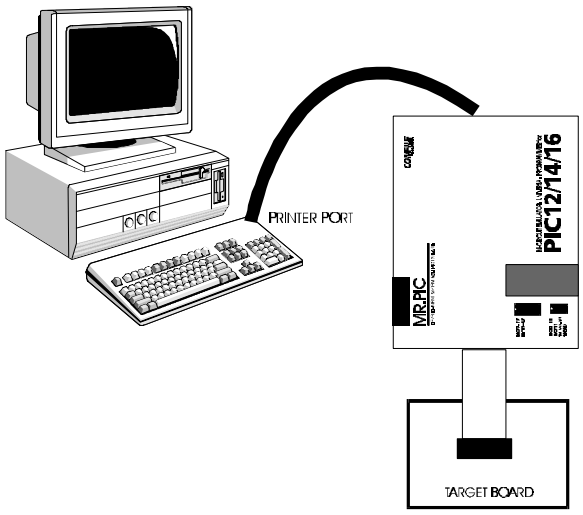


Figure 2-3. MR.PIC Hardware Installation.

Note:
MR.PIC interface the PC printer port. The cable connector of MR.PIC has 15 pin, and the other of PC has 25 pin. The cable connection is the following.

Printer port side (25pin)		MR.PIC side (15pin)
2	----------	1
3	----------	2
4	----------	3
5	----------	4
6	----------	5
10	----------	6
11	----------	7
12	----------	8
13	----------	9
15	----------	10
25	----------	15

CHAPTER 3

OPERATION

START THE MR.PIC-IDE

Now, you will step up the entire working procedure in this chapter. First double-click the MR.PIC-IDE icon.

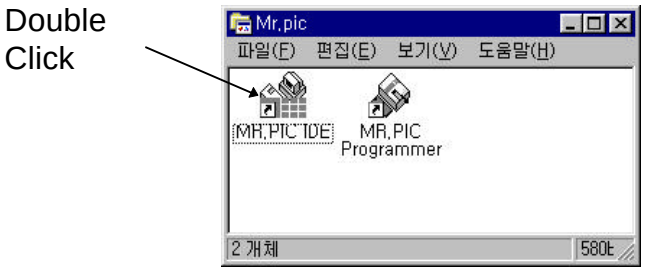
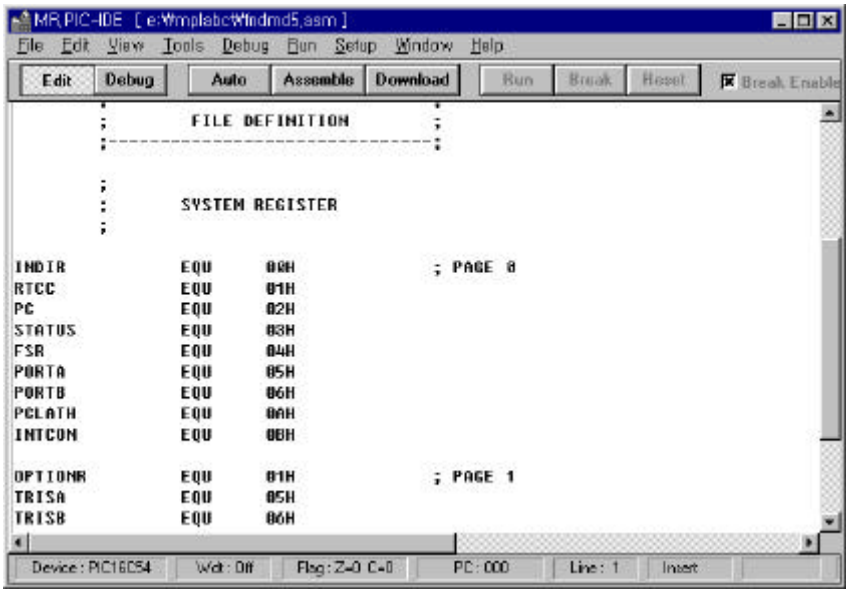


Figure 3-1. MR.PIC-IDE icon

Figure 3-2. MR.PIC-IDE initial screen

You should repeat the installation procedure again, if the preceding picture does not appear or the message of "MR.PIC Hardware Not Found !!!" appear.



In this manual, the notation for path of the menu is as <File menu/Open menu>. (It means the "Open " menu follows the "File" menu.)

OPEN

You will open the source file (.ASM file) from "Open" menu in MR.PIC-IDE. For example, "Open" the <ABC.ASM>. From now on, all works will occur in the <ABC.ASM>. (such as edit, assemble, download, run ....)

You choose the <File menu / Open menu> for open source file. When you choose "Open", the dialog box is as below.

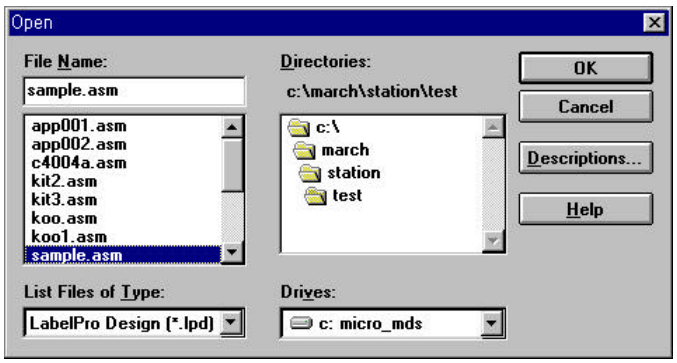


Figure 3-3. Open dialog box

EDIT

You can edit the source file, using the MR.PIC-IDE edit function. (such as Undo, Redo, Cut, Copy, Paste, Find, Replace...)

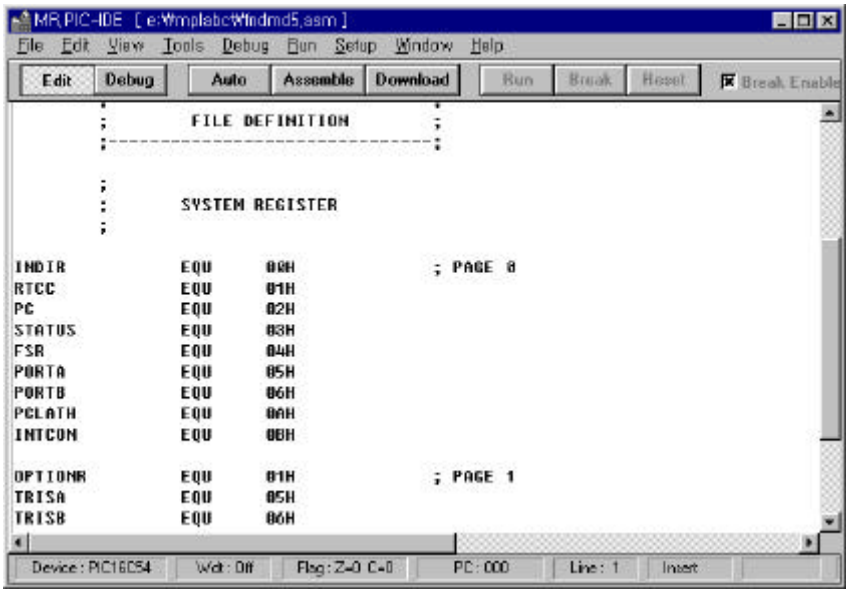


Figure 3-4. Source Editing.

ASSEMBLE

The " assemble " translates the source files into usable object files.
MR.PIC-IDE contains MPASM cross assembler for PIC microcontroller. You choose <File menu / Assemble menu >.

Note:
The created files in the Assemble are as follows.
(In case that source file is ABC.ASM)

ABC.COD	: The code file for emulator
ABC.HEX	: INTEL HEX format file (INHX8M type)
ABC.LST	: The listing file for the print.
ABC.ERR	: The error file to see the error.

The OTP programming uses the *.HEX file. HEX file is composed the INTEL HEX 8 bit format (INHX8M). You may press the assemble button on the tool bar or the F3 key for assemble.

Note :
The file is saved automatically when assemble starts

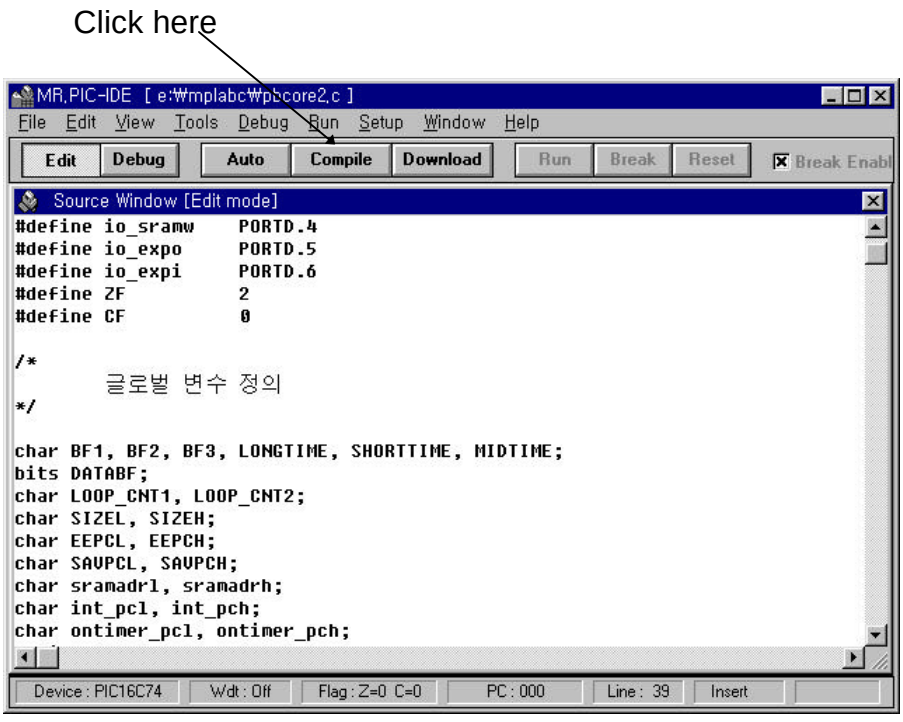


Figure 3-5. Assemble start

The Figure below shows assembling dialog box.

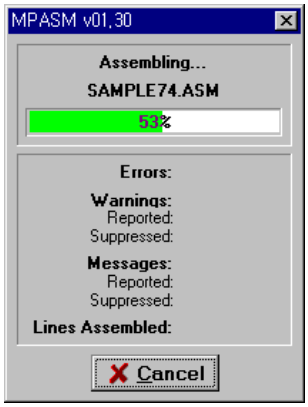


Figure 3-6. Assemble Message

The display show the error that is occupied in the assembling, the warning, and the number of the message.

Note :
When the assemble error has occurred, You can find the error using <Edit menu / Find Error menu>.

DOWNLOAD

" Download " takes the object files from assemble into MR.PIC emulation memory. To be remember, it downloads the content of the COD file into the MR.PIC emulation memory, and readjust the PC internal memory to be suitable with the present COD file. To do the download, you can choose the < File menu / Download to MR.PIC menu > or the F4 key or click < Download> button on tool bar.

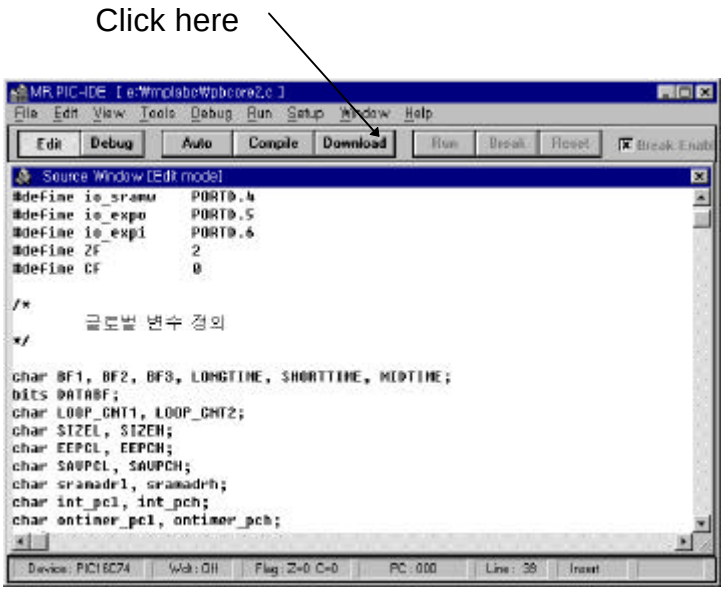


Figure 3-7. Download start

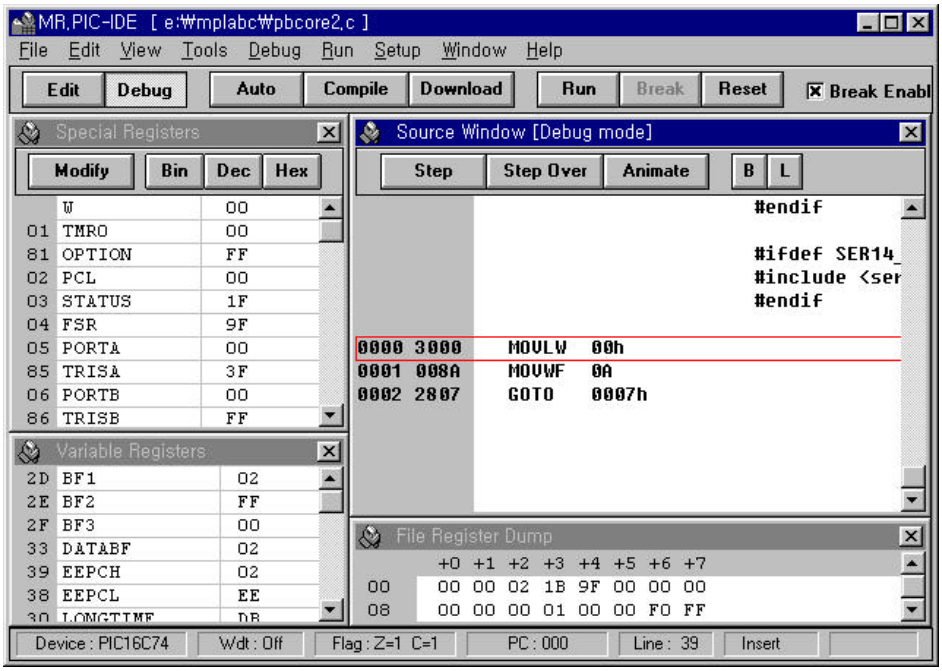


Figure 3-8. The download is completed.

When download is completed, the source file appears in the source list window. The execution bar (red box) is locates in the reset vector, the first executable address.

Note:
Like this, the environment that indicates the error and lets user to debug it is called, " Source Level debugging environment". MR.PIC-IDE supports the complete source level debugging environment. "The Execution Point" in the execution is put around the red box around the source code in the Source List window.

If the error message indicating "MR.PIC hardware not found !!!" is appears at the download procedure, you may see that MR.PIC is not connected to PC, or an oscillator is not insert, or power is down. In this case you need to check the power and the oscillator, and restart MR.PIC-IDE.

Note:
When you modify the source program, you should always run the Assemble-Download execution in order without fail. If you skip the assemble, only the state prior to the assemble runs. Therefore you must run the download after the assemble.

MR.PIC-IDE have four divided windows like below.

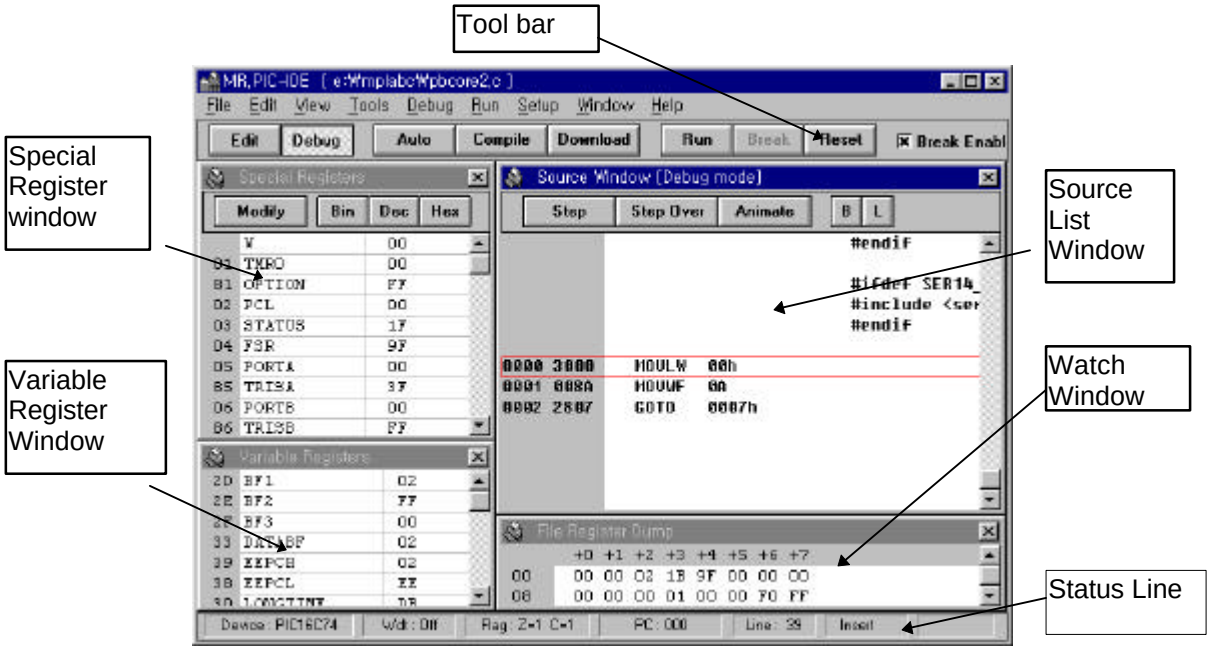


Figure 3-9. Screen setup

Window	**Description**
Title bar	Display title & source file name
Menu bar	Menu operation place.
Tool bar	Shortcut button
Special Register Window	Special function register view (such as W register, STATUS, FSR, PORTA...)
Variable Register window	User defined variable register view.
Source list window	Source list
Watch window	Register view set of user selection.
Status line	Device type, status flag...

RUN PROGRAM

You may select Run to start the MR.PIC-IDE operation. You can press the <Break> button, and stop MR.PIC-IDE operation.

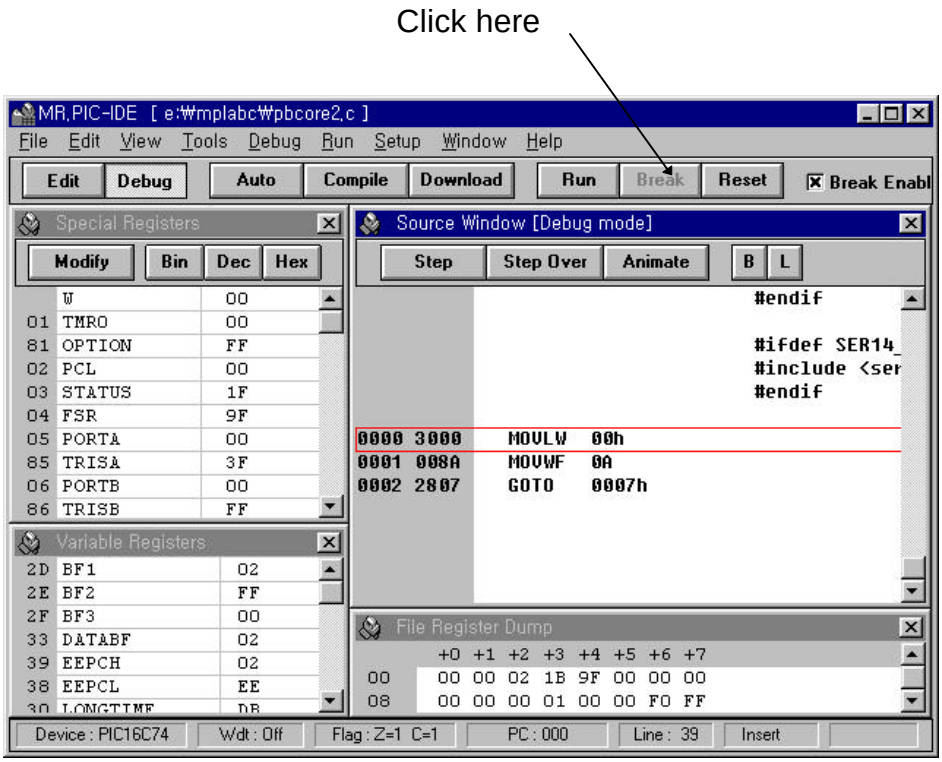


Figure 3-10. Run Start

The following Figure shows the program is running.

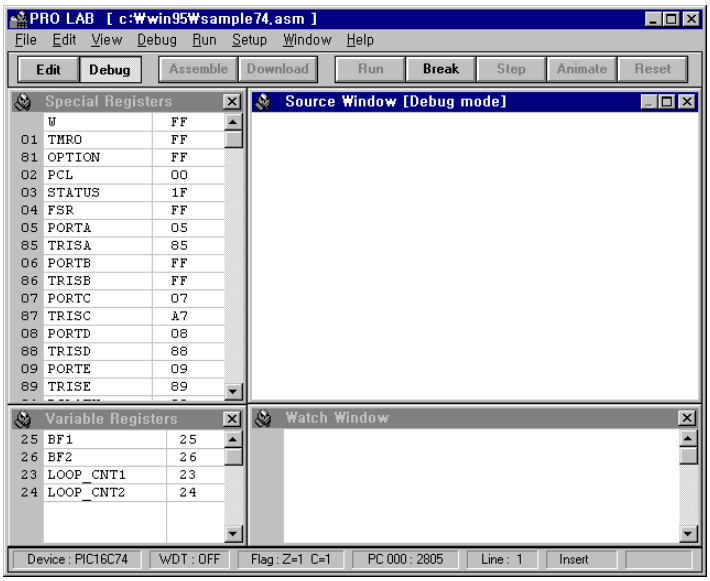


Figure 3-11. Run MR.PIC-IDE

DEBUGGING

You can use MR.PIC-IDE debugging feature to debug a program. (such as Step, BreakPoint, Modify Register...) Refer to "Chapter 5 : Debugging".

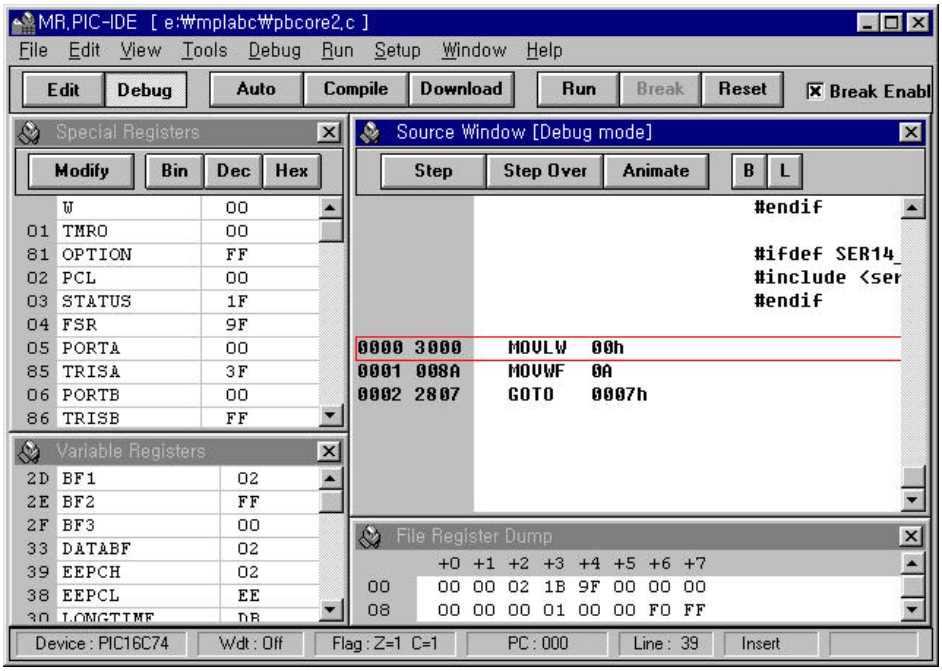


Figure 3-12. Debugging

Return to EDITOR

Selecting "Edit" button from tool-bar to switch MR.PIC-IDE Editor mode.

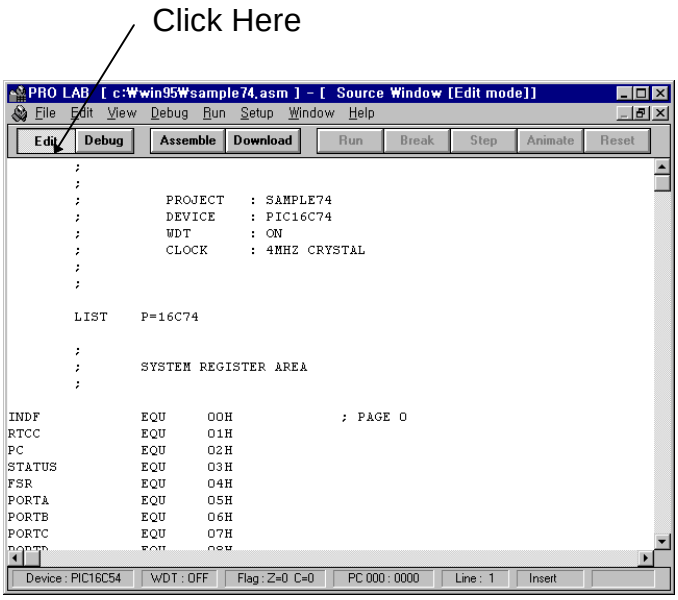
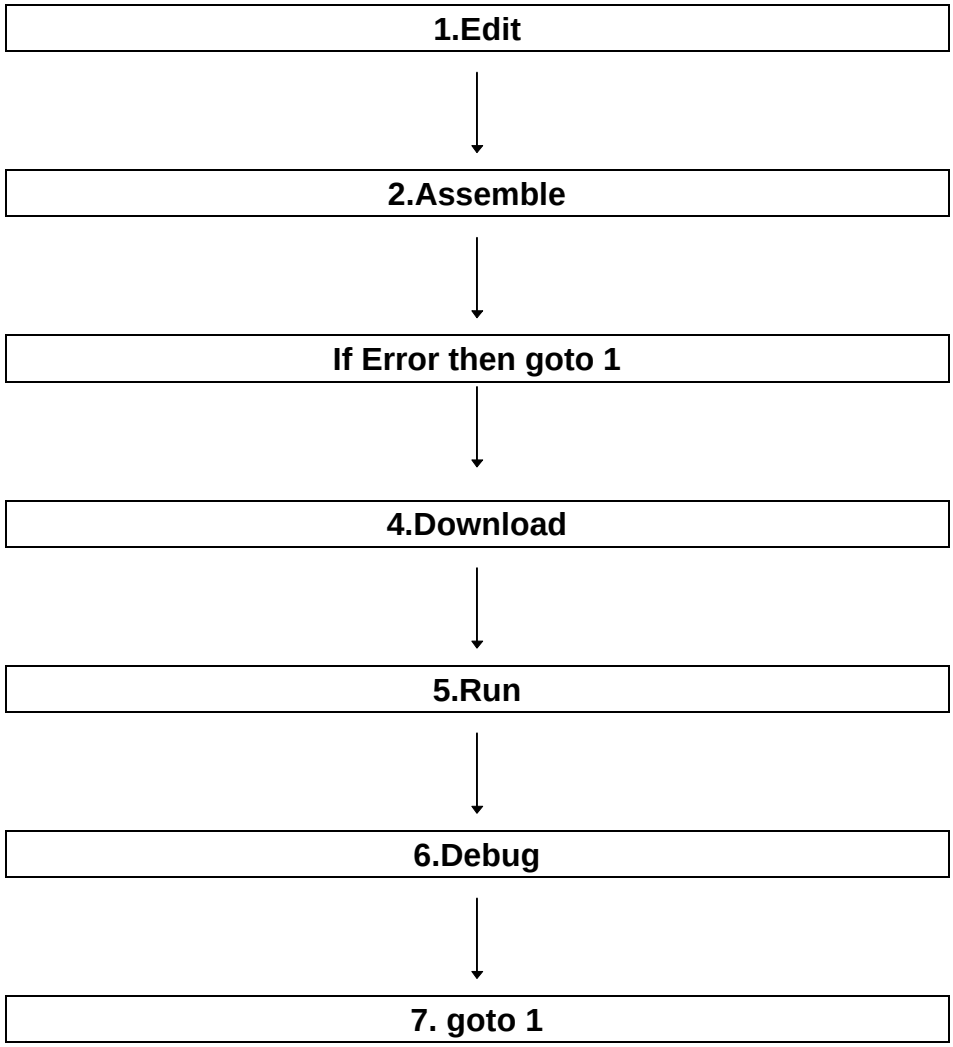


Figure 3-13. Editor Mode

THE SUMMARY

Well, we can digest the MR.PIC-IDE emulator work like the following.



CHAPTER 4

SUMMARY OF MENUS

FILE MENU

Open...

Detailed description is in the previous chapter. This menu open the source file.

Assemble...

Detailed description is in the previous chapter. This menu assemble the source file determined in the <Open Project>, and makes the an executable object file. It uses MPASM assembler, and the output file for MR.PIC-IDE download is COD files.

Download to Firmware

This menu downloads the COD file to MR.PIC emulation memory.

Save Configuration & Screen Setup

This menu save system configuration (such as Printer port address..) and screen setup to MR.PIC-IDE.INI file.

EDIT MENU

Find Error

When an assemble error is occurred, then a red bar appeared. You can find the error by this menu.

VIEW MENU

Source Window

This menu displays the Source list.

File Register Dump

This menu displays the File Register Dump.

Special Function Registers

This menu displays the Special Function Registers.

Variable Register

This menu displays the Variable Registers.

Watch Window

This menu displays the Watch window.

Program Memory Usage Map

This menu displays the Program Memory Usage Map.

Program Memory Dump

This menu displays the Program Memory Dump.

Error file

This menu displays the Error file (*.ERR)

Listing file

This menu displays the Listing file (*.LST)

Header file or Referance file

This menu displays the Header file (*.H, *.INC) or Referance file.

DEBUG MENU

Clear All Break Point

This menu clears all break-points on source list window. Reference the following chapter for the detailed explanation about the break-point.

Clear All Watch

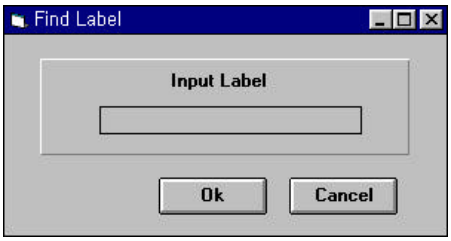
This menu clears all watch setup on watch window.

Find Break Point

This menu find break-points on source list window.

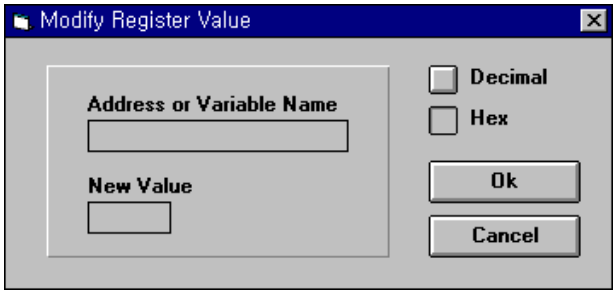
Find Label...

This menu find label, you can see the dialog box as follows.



Modify Register Value...

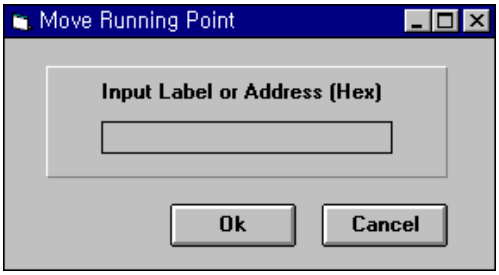
This menu changes the present register value, you can see the dialog box as follows.



You can input the address (or Variable Name) and the new value. When the value gets out of range, "Invalid Range" error comes out.

Modify Running Point...

This menu moves the present running point. When you choose it, the following dialog box follows.



You can input the address or label. Address must be input hex value. (Such as 0F0, 1A, 0AA....)

Edit Mode

This menu switch to Editor Mode.

Debug Mode

This menu switch to Debug Mode.

RUN MENU

Run

This menu execute the program.

Break

This menu stop the program.

Single Step

This menu make program to run step-by-step.

Animate

This menu make program to slow run. You can adjust animate speed using <Setup menu / Animate speed Setting... menu>.

Reset

This menu reset the processor.

SETUP MENU

Enable Watch Dog Timer

This menu enables watch dog timer, and reset the chip at once.

Disable Watch Dog Timer

This menu disables watch dog timer, and reset the chip at once.

Animate Speed Setting...

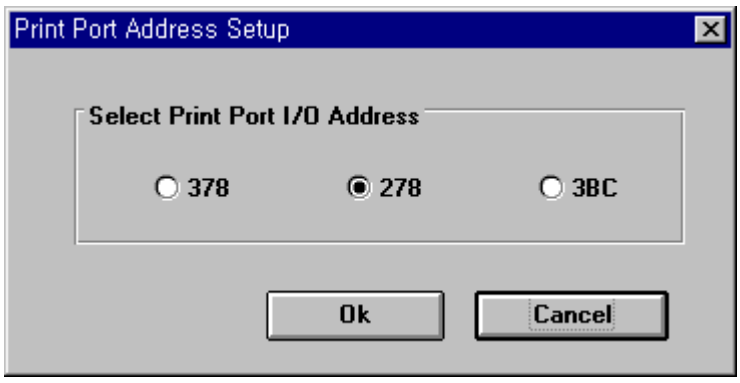
This menu adjust the Anmate Speed.

Radix Setting...

This menu is set the display radix. (for variable register window and special register window)

Printer Port Address...

This menu is Printer port address setup menu. When you choose it, the following dialog box follows.



You can find <Printer port address> using this menu. After find, click <File menu / Save Configuration & Screen setup menu> for save information.

Note :
If a massage "MR.PIC hardware not found !!!" appears when booting MR.PIC-IDE. You first check out if your PRINT PORT port address is OK.

CHAPTER 5

DEBUGGING

BREAK POINT

Break point is a very important function for the microcontroller development. It gives you the function to stop the execution at the wanted point while running and you can check the status of program execution.

MR.PIC-IDE gives you the unlimited number of Break-Point in this function. It means you can set Break-Point covering the total program memory. To set Break-Point it, double click the wanted point on < Source List Window >. The following display shows the Break-Point set.

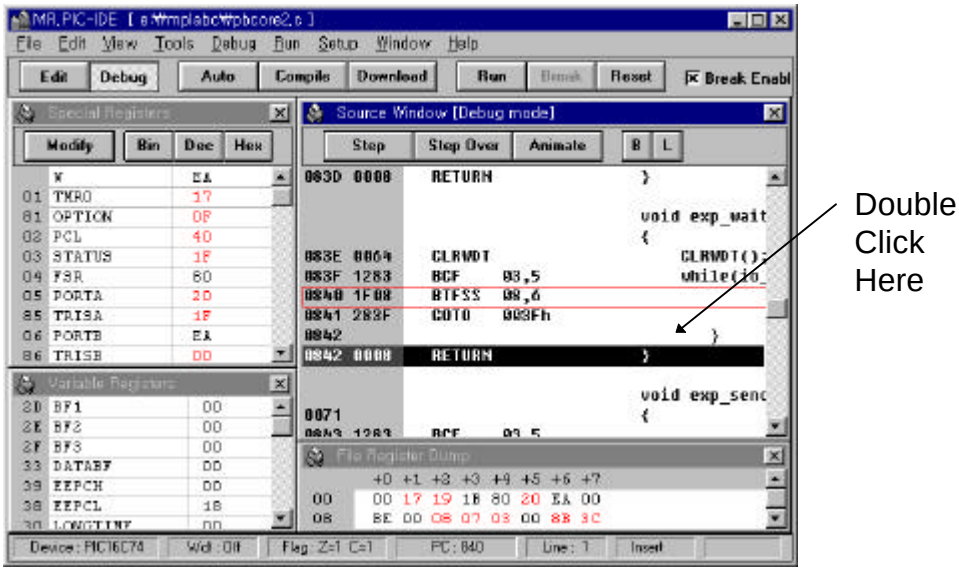


Figure 5-1. Break point set

Chosen Break-Point is shown in the reverse bar state. If you double click the same point again, the it toggles back to normal.

To cancel the present setups of all Break-Points, you can use the <Debug menu / Clear All Break Point menu >.

MOVE RUNNING POINT

Running point means the present address on activity. Running point appears the red box in < Source List window >. If you move this, you locate the cursor at wanted point and click the mouse right button. Then you can see the move of the running point.

Note :
You move running point, when it gets out of the loop or you want to start it at specified point.

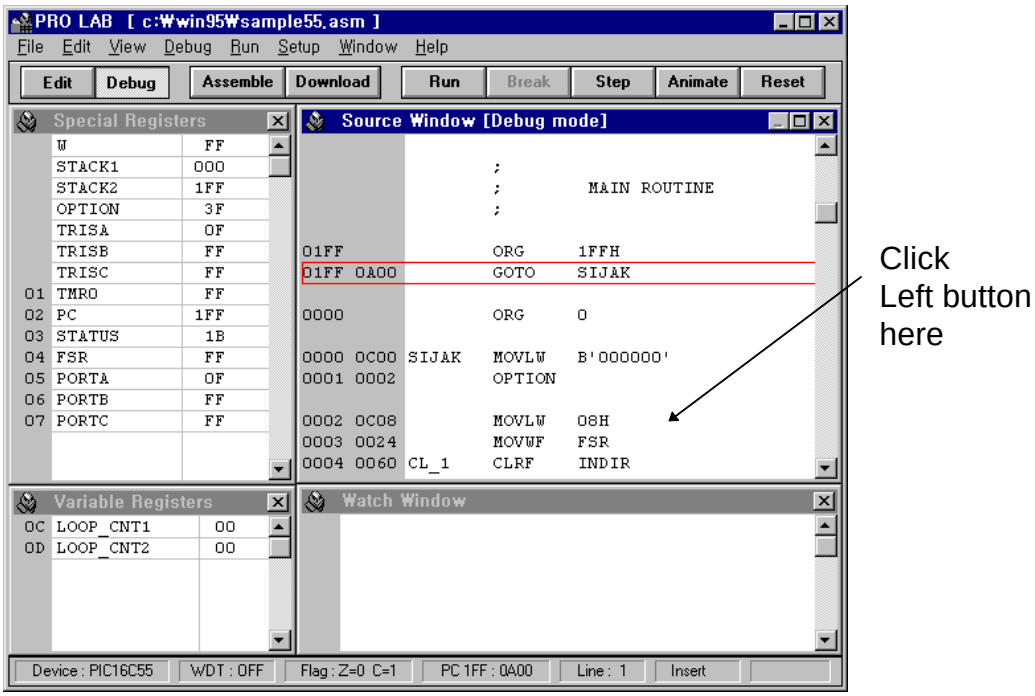


Figure 5-3. Move running point.

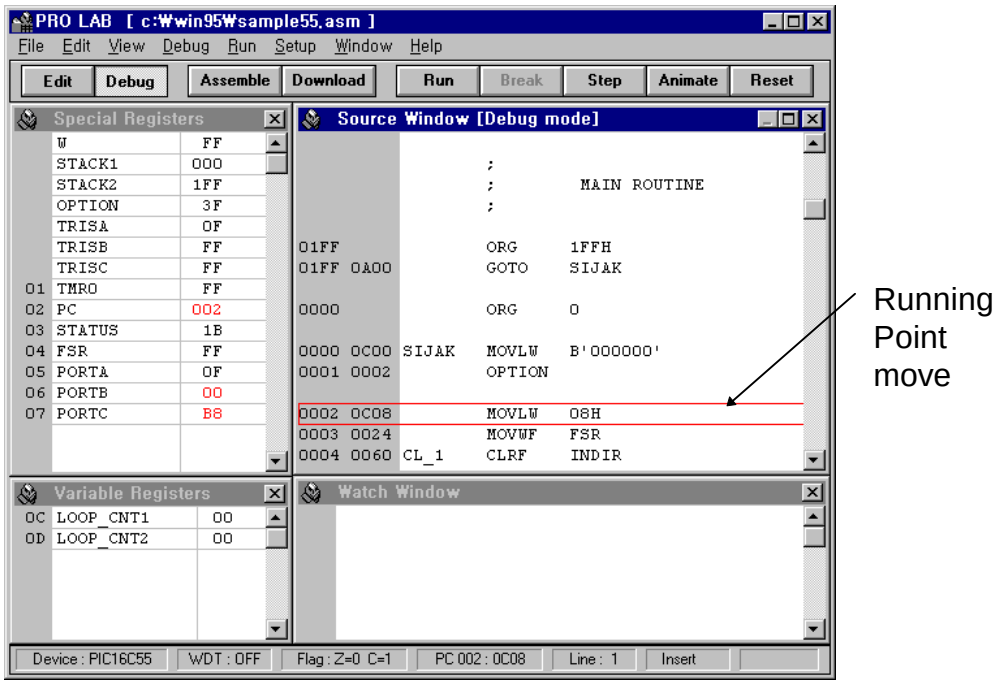
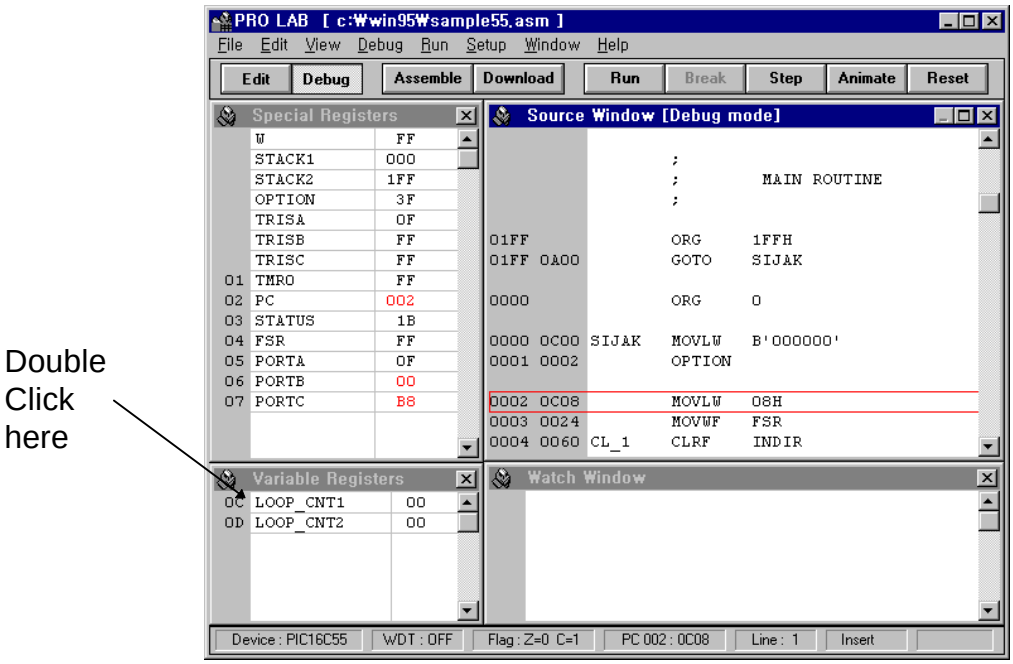


Figure 5-4. Move running point.

CHANGE THE REGISTER VALUE

You can change the register value by two kinds of method. The one is to choose < Debug menu / Modify Register Value menu >, and the other is to double click the target point in "Variable window" or "Watch window"



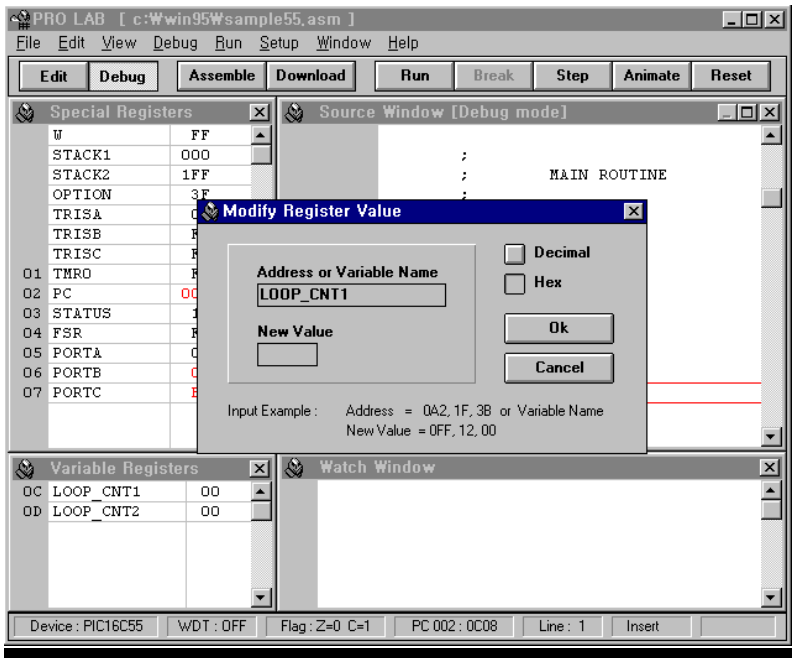


Figure 5-5. Modify register

VARIABLE REGISTER WINDOW

Variable register window has very special function. It shows only symbol that user defined in general purpose area. Variable window displays the VARIABLE name that you have defined with the VARIABLE command in your source file.

"VARIABLE" means the register that a user defines its name. Suppose that you define as follows in the Source file.

```
VARIABLE    BF1         = 25H
VARIABLE    BF2         = 26H
VARIABLE    LOOP_CNT1   = 23H
VARIABLE    LOOP_CNT2   = 24H
```

LOOP_CNT1 and LOOP_CNT2 are what is called "Variable". Variables are always locateds in <Variable Register window> so that you can easily check on work.

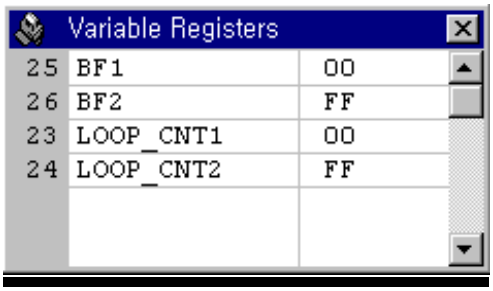


Figure 5-7. Variable window

WATCH WINDOW

Watch window has a very useful function to verify register value. To use watch window just click right mouse button where you want to use in "Variable register window" or "Special register window". Then that is added in watch window.

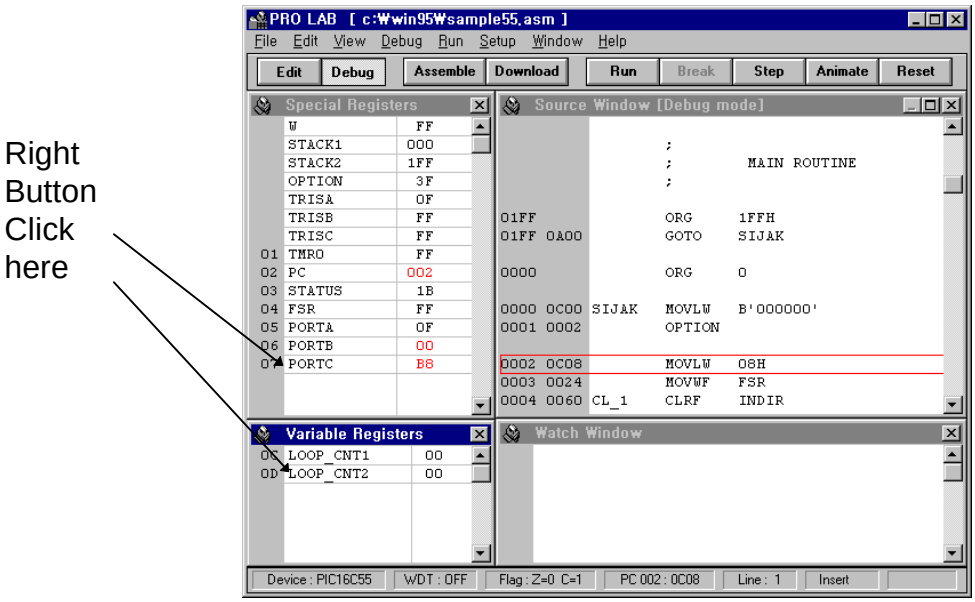


Figure 5-8. Watch window

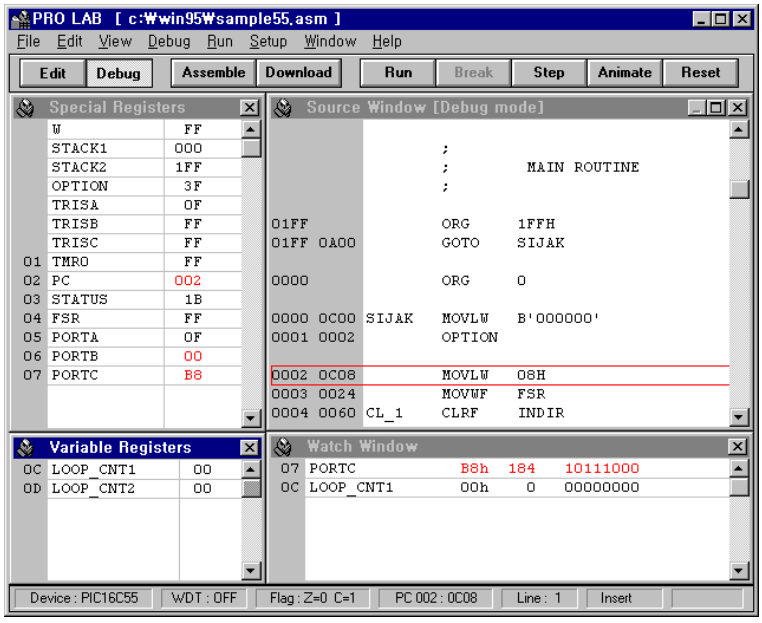


Figure 5-9. Watch window

If you have anything you want to delete in watch window, move your mouse pointer to wanted point, and click right mouse button.

APPENDIX

Sample program :
PIC16C74 A/D conversion

```
;
;       PROJECT     : SAMPLE74
;       DEVICE      : PIC16C74
;       WDT         : ON
;       CLOCK       : 4MHZ CRYSTAL
;
;


        LIST    P=16C74

       ;
       ;       SYSTEM REGISTER AREA
       ;

INDF                EQU         00H             ; PAGE 0
RTCC                EQU         01H
PC                  EQU         02H
STATUS              EQU         03H
FSR                 EQU         04H
PORTA               EQU         05H
PORTB               EQU         06H
PORTC               EQU         07H
PORTD               EQU         08H
PORTE               EQU         09H
PCLATH              EQU         0AH
INTCON              EQU         0BH
PIR1                EQU         0CH
PIR2                EQU         0DH
TMR1L               EQU         0EH
TMR1H               EQU         0FH
T1CON               EQU         10H
TMR2                EQU         11H
T2CON               EQU         12H
SSPBUF              EQU         13H
SSPCON              EQU         14H
CCPR1L              EQU         15H
CCPRLH              EQU         16H
CCP1CON             EQU         17H
RCSTA               EQU         18H
TXREG               EQU         19H
RCREG               EQU         1AH
CCPR2L              EQU         1BH
CCPR2H              EQU         1CH
CCP2CON             EQU         1DH
ADRES               EQU         1EH
ADCON0              EQU         1FH

OPTIONR             EQU         01H             ; PAGE 1
TRISA               EQU         05H
TRISB               EQU         06H
TRISC               EQU         07H
TRISD               EQU         08H
TRISE               EQU         09H
PIE1                EQU         0CH
PIE2                EQU         0DH
```

```
PCON                EQU         0EH
PR2                 EQU         12H
SSPADD              EQU         13H
SSPSTAT             EQU         14H
TXSTA               EQU         18H
SPBRG               EQU         19H
ADCON1              EQU         1FH

       ;
       ;       USER DEFINE AREA
       ;

VARIABLE            LOOP_CNT1   = 23H
VARIABLE            LOOP_CNT2   = 24H
VARIABLE            BF1         = 25H
VARIABLE            BF2         = 26H

       ;
       ;       BIT DEFINITION
       ;

CF                  EQU         .0          ; STATUS
DC                  EQU         .1
ZF                  EQU         .2
PD                  EQU         .3
TO                  EQU         .4
RP0                 EQU         .5

GO_DONE             EQU         .2          ; ADCON0

       ;
       ;    RESET & INTERRUPT VECTOR
       ;

       ORG          0
       GOTO         SIJAK

       ;
       ;        START ROUTINE
       ;
       ORG          5
SIJAK
       BSF          STATUS,RP0              ; PAGE1
       MOVLW        B'00011111'
       MOVWF        TRISA
       MOVLW        B'00000000'
       MOVWF        TRISB
       MOVLW        B'00000000'
       MOVWF        TRISC
       MOVLW        B'00000000'
       MOVWF        TRISD
       MOVWF        TRISE
       MOVLW        B'00000101'
       MOVWF        OPTIONR

       MOVLW        B'00000010'
       MOVWF        ADCON1

       BCF          STATUS,RP0
       MOVLW        B'11000001'
       MOVWF        ADCON0

       MOVLW        20H         ; CLEAR RAM (20H~07FH)
       MOVWF        FSR
```

```
CLR_1   CLRF        INDF
        INCF        FSR
        BTFSS       FSR,7
        GOTO        CLR_1

        MOVLW       0A0H        ; CLEAR RAM (0A0H~0FFH)
        MOVWF       FSR
CLR_2   CLRF        INDF
        INCFSZ      FSR
        GOTO        CLR_2

MAIN_LOOP
        MOVLW       .10
        CALL        DELAY_US
        BSF         ADCON0,GO_DONE
KI_1    NOP
        BTFSC       ADCON0,GO_DONE
        GOTO        KI_1
        MOVF        ADRES,W
        MOVWF       BF1
        MOVWF       PORTB       ; RESULT OUTPUT
        MOVWF       PORTC
        MOVWF       PORTD
        GOTO        MAIN_LOOP

DELAY_US                        ; DELAY FOR SAMPLE & HOLD

        MOVWF       LOOP_CNT1
DL_1    NOP
        DECFSZ      LOOP_CNT1
        GOTO        DL_1
        RETURN

        END
```

THANK YOU !

If problem is occur when you install your system or you use this tool, please contact to COMFILE TECHNOLOGY. TEL 82-2-711-2592, FAX 82-2-711-2593 or E-MAIL comfile@comfile.co.kr.
Thank you !!!

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